

Sushant

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Aspiring Tech Innovator | Electrical Engineering Student | Hardware & VLSI Enthusiast

Electrical Engineering student at Delhi Technological University with a deep passion for semiconductor engineering, VLSI design, and hardware innovation. Experienced in rapid prototyping with Arduino and FPGA, and excited by the intersection of AI/ML with chip design and intelligent hardware acceleration. Proven ability to bridge the gap between custom hardware RTL architectures and full-stack software integration.

Education

Delhi Technological University (Formerly DCE)

B.Tech Electrical Engineering

New Delhi, India

Jul 2025 – Jul 2029

Kendriya Vidyalaya

Secondary & Senior Secondary Education

India

Mar 2012 – Mar 2024

Experience

Houston International Foods Pvt. Ltd.

Full-Stack Web Developer Intern

Mumbai, Maharashtra

Jun 2025 – Jul 2025

- Developed and deployed web applications for the Information Technology department, streamlining internal data flows.
- Executed frontend-backend architecture design, demonstrating strong problem-solving skills and clean code execution.
- Recommended by the organization for future opportunities in website development due to punctual delivery and collaborative teamwork.

Projects

FPGA-Accelerated Edge Intelligence System | *PYNQ-Z2 (Zynq-7020), ESP32-S3, VHDL, C, UART* SemiXthon 2026

- Engineered a Tiny Edge AI Accelerator capable of real-time anomaly detection and pattern recognition at the edge with zero cloud dependency.
- Designed a custom RTL processing pipeline (Threshold Detector, FSM Classifier, Pattern Matcher) that achieved a verified per-frame latency of $57.17 \mu s$, operating 87x faster than the required 5ms target.
- Established a high-speed (921600 baud) UART communication pipeline between the ESP32 sensor host and the FPGA processing core.

Advanced Driver Assistance Systems (ADAS) | *Python, OpenCV, PyTorch, C++, Sensors*

- Architected an integrated ADAS pipeline focused on real-time data processing and computer vision.
- Fused hardware sensor data with machine learning object detection models to evaluate real-time environmental conditions and reduce processing latency.

Hand-Gesture Recognition via Custom RTL Accelerator | *FPGA, Verilog/VHDL, Python*

- Designed and implemented a Custom Register-Transfer Level (RTL) architecture specifically optimized for edge AI logic.
- Shifted traditional software-based AI recognition to a dedicated hardware accelerator, drastically reducing power consumption and inference time.

Hackathons & Achievements

AM Hacks 2.0 (AssetMerkle IGDTUW)

Top 20 Finalist (Team SJJ)

New Delhi, India

Feb 2025

- Ranked in the Top 20 out of all competing teams in a highly competitive 48-hour offline hackathon.
- Served as the sole technical lead for the team, orchestrating rapid development and clean execution.
- Architected and built the entire prototype web app (exclusively software/frontend UI) during a major strategic pivot, demonstrating high-pressure problem-solving capabilities.

Technical Skills

Hardware Platforms: FPGA (PYNQ-Z2, Boolean), ESP32, Arduino

Semiconductor & VLSI: CMOS Logic Design, MOSFET Device Physics, VLSI Design, Verilog, RTL.

Programming Languages: C/C++, Python

AI/ML: AI-driven Hardware Acceleration, Automated Layout Optimization, TinyML/Edge AI, PyTorch, Scikit-learn, NumPy

Core Concepts: Digital Logic Design, Embedded Systems, Rapid Prototyping, Data Structures & Algorithms